Mathematics for Cryptography II: Security Analysis and Polynomials

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github.com/BlockstreamResearch/zk-book



Plan

1 Quick Recap

2 Security Analysis

3 Polynomials

What will we learn today?

- How to define the security formally.
- How to prove that a construction is secure: security reductions.
- How to read. . . This. . .

Definition 4 (Hiding Commitment)**.** *A commitment scheme is said to be hiding if for all PPT adversaries $\mathcal{A}$ there exists a negligible function $\mu(\lambda)$ such that.*

$$\left| \mathrm{P}\left[b = b' \;\middle|\; \begin{array}{l} \mathrm{pp} \leftarrow \mathrm{Setup}(1^\lambda); \\ (x_0, x_1) \in \mathsf{M}_{\mathrm{pp}}^2 \leftarrow \mathcal{A}(\mathrm{pp}), b \xleftarrow{\$} \{0,1\}, r \xleftarrow{\$} \mathsf{R}_{\mathrm{pp}}, \\ \mathbf{com} = \mathrm{Com}(x_b; r), b' \leftarrow \mathcal{A}(\mathrm{pp}, \mathbf{com}) \end{array} \right] - \frac{1}{2} \right| \leqslant \mu(\lambda)$$

where the probability is over $b, r,$ Setup *and $\mathcal{A}$. If $\mu(\lambda) = 0$ then we say the scheme is* perfectly hiding.

Definition 5 (Binding Commitment)**.** *A commitment scheme is said to be binding if for all PPT adversaries $\mathcal{A}$ there exists a negligible function μ such that.*

$$\mathrm{P}\left[\mathrm{Com}(x_0; r_0) = \mathrm{Com}(x_1; r_1) \wedge x_0 \neq x_1 \;\middle|\; \begin{array}{l} \mathrm{pp} \leftarrow \mathrm{Setup}(1^\lambda), \\ x_0, x_1, r_0, r_1 \leftarrow \mathcal{A}(\mathrm{pp}) \end{array} \right] \leqslant \mu(\lambda)$$

where the probability is over Setup *and $\mathcal{A}$. If $\mu(\lambda) = 0$ then we say the scheme is* perfectly binding.

Figure: This is not that hard as it seems. Figure from "*Bulletproofs: Short Proofs for Confidential Transactions and More*"

Quick Recap

Quick Recap

1. We know how to read formal statements, like

$$(\forall n \in \mathbb{N})\,(\exists k \in \mathbb{Z}) : \{n = 2k + 1 \vee n = 2k\}$$

2. Group $\mathbb{G}$ is a set with a binary operation that satisfies certain rules. In this lecture, we will use the **multiplicative** notation: for example, g^{α} means g multiplied by itself α times.
3. Probability of event E is denoted by $\Pr[E]$, and $x \leftarrow\!\$\ S$ means that x is sampled uniformly at random from S.
4. $\#S$ is the number of elements in the finite set S.

Security Analysis

Introducing the Cipher: a bit of Formalism

We will consider an example of a cipher to demonstrate the notion of security.

Let us introduce three sets:

- $\mathcal{K}$ — a set of all possible keys.
- $\mathcal{M}$ — a set of all possible messages. For example, $\mathcal{M} = \{0,1\}^n$ — all binary strings of length n.
- $\mathcal{C}$ — a set of all possible ciphertexts.

The cipher is defined over a tuple $(\mathcal{K}, \mathcal{M}, \mathcal{C})$.

Tiny Note

Cryptography is a very formalized field, but everything considered is well-known to you.

Introducing the Cipher

Definition

Cipher $\mathcal{E} = (E, D)$ over the space $(\mathcal{K}, \mathcal{M}, \mathcal{C})$ consists of two efficiently computable methods:

- $E : \mathcal{K} \times \mathcal{M} \to \mathcal{C}$ — encryption method, that based on the provided message $m \in \mathcal{M}$ and key $k \in \mathcal{K}$ outputs the cipher $c = E(k, m) \in \mathcal{C}$.
- $D : \mathcal{K} \times \mathcal{C} \to \mathcal{M}$ — decryption method, that based on the provided cipher $c \in \mathcal{C}$ and key $k \in \mathcal{K}$ outputs the message $m = D(k, c) \in \mathcal{M}$.

We require the **correctness**:

$$D(k, E(k, m)) = m \quad \text{// for all messages } m \in \mathcal{M} \text{ and all keys } k \in \mathcal{K}.$$

Question

What else cipher must have to be practical?

Defining Security

Typically, the security is defined as a **game** between the adversary ($\mathcal{A}$) and the challenger ($\mathcal{C}h$).

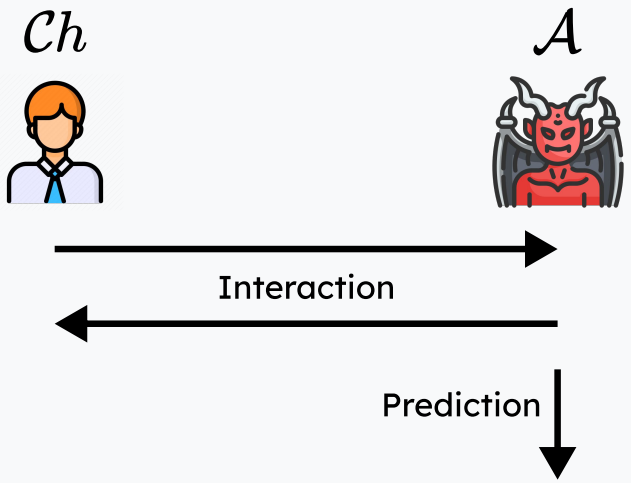


Figure: Challenger $\mathcal{C}h$ follows a straightforward protocol, while the adversary $\mathcal{A}$ might take any (efficient) strategy to win the game.

Efficient

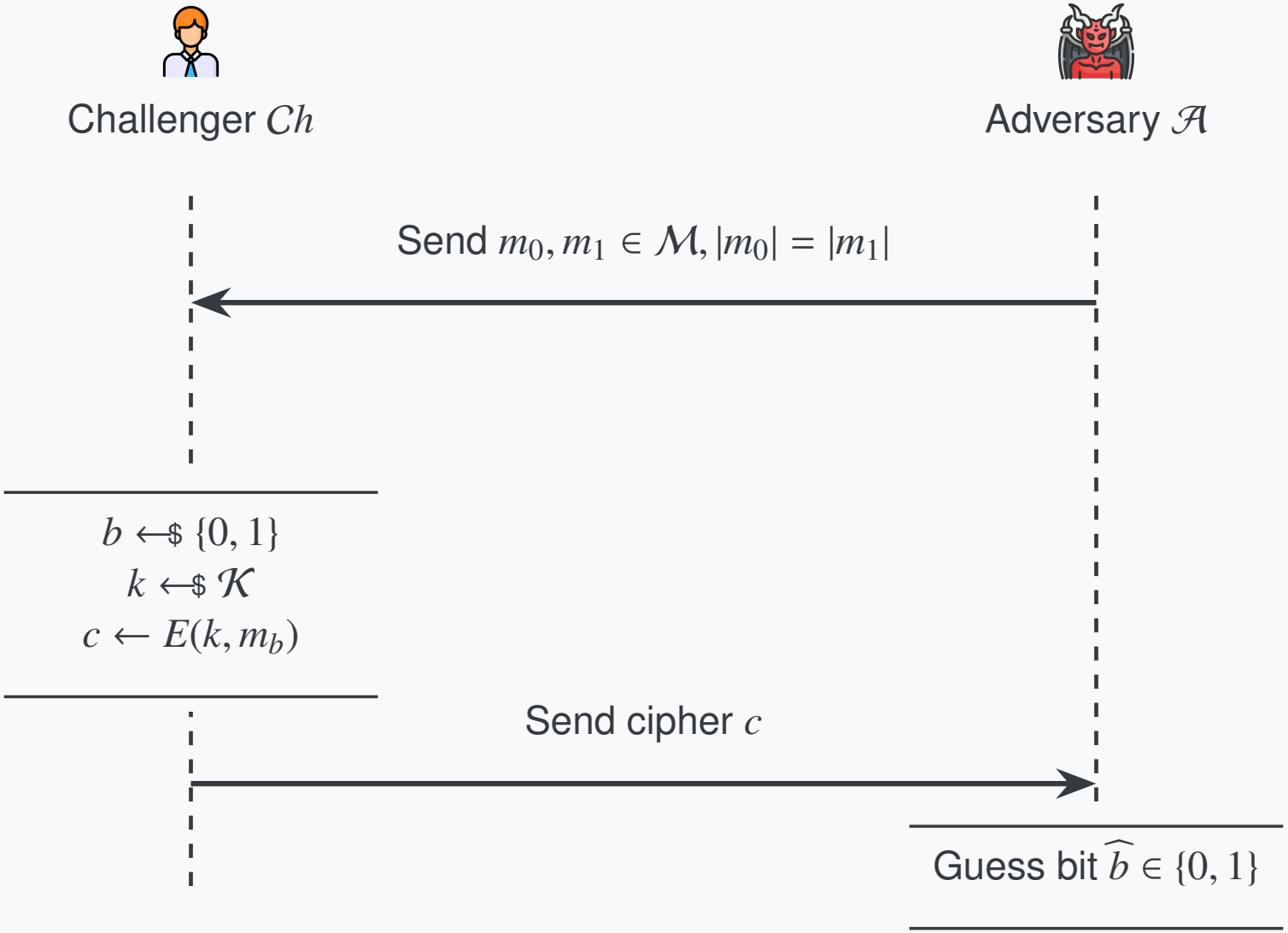
Efficient means that an algorithm requires polynomial time and memory with respect to the security parameter (defined in a moment).

Semantic Security: Bit Guessing Game

Let us play the following game between adversary $\mathcal{A}$ and challenger $\mathcal{C}h$:

1: $\mathcal{A}$ picks two messages $m_0, m_1 \in \mathcal{M}$ of the same length on his choice.
2: $\mathcal{C}h$ picks a random key $k \leftarrow\$ \mathcal{K}$ and random bit $b \leftarrow\$ \{0, 1\}$ and sends the cipher $c = E(k, m_b)$ to $\mathcal{A}$.
3: $\mathcal{A}$ is trying to guess the bit b by using the cipher c.
4: $\mathcal{A}$ outputs the guess $\widehat{b}$.

Semantic Security: Bit Guessing Game



Semantic Security: Advantage

Question #1

Suppose our cipher is perfect. What is the probability that the adversary guesses the bit b correctly? (that is, $b = \widehat{b}$)

Definition

The **advantage** of the adversary $\mathcal{A}$ in the bit guessing game against the cipher $\mathcal{E}$ is

$$\mathrm{Adv}^{\mathrm{SS}}_{\mathcal{A},\mathcal{E}} := \left| \Pr[\widehat{b} = b] - \frac{1}{2} \right| .$$

Definition

The cipher $\mathcal{E}$ is called **semantically secure** if for any efficient adversary $\mathcal{A}$ the advantage $\mathrm{Adv}^{\mathrm{SS}}_{\mathcal{A},\mathcal{E}}$ is **negligible**.

Advantage

Question #2

If adversary can guess the bit with probability 0.000000001, is the cipher semantically secure?

Question #3

If adversary has the advantage 0.0001, is the cipher semantically secure?

Note

Advantage is just a measure of how much information is leaked to the adversary. The smaller the advantage, the better the cryptographic system. Formally, we want advantage to be **negligible**.

But what does it mean to be **negligible**?

Negligibility

In practice, negligible means below 2^{-128} (called 128 bits of security).

In theory, however...

Definition

Security parameter, denoted by $\lambda \in \mathbb{N}$, is the "logarithmic size" of the scheme. We then treat the advantage as a function $\mathsf{Adv}^{\mathsf{SS}}_{\mathcal{A},\mathcal{E}}(\lambda) : \mathbb{N} \to \mathbb{R}_{\geq 0}$.

Example

λ might correspond to the logarithm of the order of the underlying finite field or, for a cipher, we might set $\lambda :=$ "ciphertext size in bits".

Negligible Functions

Now, the probability of adversary winning the game depends on λ and we want this dependence to decrease rapidly.

Definition

A function $f : \mathbb{N} \to \mathbb{R}$ is called **negligible** if for all $\gamma \in \mathbb{R}_{>0}$ there exists $n_\gamma \in \mathbb{N}$ such that for any $n \geq n_\gamma$ we have $|f(n)| < 1/n^\gamma$.

Less formally: f decays faster than any polynomial. The alternative definition, which is probably easier to interpret, is the following.

Theorem

A function $f : \mathbb{N} \to \mathbb{R}$ is **negligible** if and only if for any $\gamma \in \mathbb{R}_{>0}$, we have

$$\lim_{\lambda \to \infty} f(\lambda)\lambda^\gamma = 0.$$

Negligible Functions Examples

Example

The function $f(\lambda) = 2^{-\lambda}$ is negligible since for any $\gamma \in \mathbb{R}_{>0}$ we have $\lim_{\lambda\to\infty} 2^{-\lambda}\lambda^{\gamma} = 0$.
The function $g(\lambda) = \frac{1}{\lambda!}$ is also negligible for similar reasons.

Example

The function $h(\lambda) = \frac{1}{\lambda}$ is not negligible since for $\gamma = 1$ we have

$$\lim_{\lambda\to\infty} \frac{1}{\lambda} \times \lambda = 1 \neq 0.$$

Question

Is the function $u(\lambda) = \lambda^{-10000}$ negligible?

Semantic Security in One Line

Many cryptographic papers avoid drawing the game explicitly. Here is another way to write the same security definition.

Definition

The encryption scheme $\mathcal{E}$ with a security parameter $\lambda \in \mathbb{N}$ is **semantically secure** if for any efficient adversary $\mathcal{A}$ we have:

$$\left| \Pr\left[b = \widehat{b} \;\middle|\; \begin{array}{l} m_0, m_1 \leftarrow \mathcal{A}(\cdot) \\ k \leftarrow\!\!\$\ \mathcal{K} \\ c_i \leftarrow E(k, m_i), i \in \{0,1\} \\ b \leftarrow\!\!\$\ \{0,1\}, \widehat{b} \leftarrow \mathcal{A}(c_b) \end{array} \right] - \frac{1}{2} \right| = \mathsf{negl}(\lambda).$$

How to read it

$\Pr[A \mid B]$ is "the probability of A, given that B occurred". On the right, we list the steps of the game; on the left, the event we are interested in.

Message Recovery Game

Any reasonable notion of security should rule out recovering the message from the ciphertext with probability better than random guessing, which is $1/\#\mathcal{M}$.

1: Ch computes $m \leftarrow\$ \mathcal{M}$, $k \leftarrow\$ \mathcal{K}$, $c \leftarrow E(k, m)$ and sends c to $\mathcal{A}$.
2: $\mathcal{A}$ outputs a message $\widehat{m} \in \mathcal{M}$.

Definition (Security against message recovery)

Define $\mathsf{Adv}^{\mathsf{MR}}_{\mathcal{A},\mathcal{E}}(\lambda) := \left|\Pr[\widehat{m} = m] - \frac{1}{\#\mathcal{M}}\right|$. A cipher $\mathcal{E}$ is **secure against message recovery** if for all efficient adversaries $\mathcal{A}$, $\mathsf{Adv}^{\mathsf{MR}}_{\mathcal{A},\mathcal{E}}(\lambda) = \mathsf{negl}(\lambda)$.

Theorem

If $\mathcal{E}$ is semantically secure, then $\mathcal{E}$ is secure against message recovery.

The Key Idea of a Reduction

Let A := "$\mathcal{E}$ is semantically secure" and B := "$\mathcal{E}$ is secure against message recovery". We want A $\Rightarrow$ B.

- Both statements are about *all* efficient adversaries: hard to prove directly.
- Their negations need just *one* adversary with a non-negligible advantage.

$$(\mathrm{A} \Rightarrow \mathrm{B}) \iff (\neg\mathrm{B} \Rightarrow \neg\mathrm{A}).$$

So: given an efficient adversary $\mathcal{A}$ breaking message recovery, we build an adversary $\mathcal{B}$ breaking semantic security. $\mathcal{B}$ uses $\mathcal{A}$ as a black box (*oracle*), written $\mathcal{B}^{\mathcal{A}}$.

Constructing the Reduction

1: $\mathcal{B}$ generates $m_0, m_1 \in \mathcal{M}$ and sends them to the semantic security challenger.
2: The challenger sends a ciphertext c to $\mathcal{B}$. Since $\mathcal{B}$ can use the message recovery oracle, it sends c to $\mathcal{A}$.
3: $\mathcal{A}$ returns $\widehat{m}$. If $\widehat{m} = m_0$, then $\mathcal{B}$ sets $\widehat{b} := 0$; otherwise, $\widehat{b} := 1$.
4: $\mathcal{B}$ sends $\widehat{b}$ to the semantic security challenger.

One can show that $\mathrm{Adv}^{\mathrm{SS}}_{\mathcal{B},\mathcal{E}} = \mathrm{Adv}^{\mathrm{MR}}_{\mathcal{A},\mathcal{E}}$, so $\mathcal{B}$ breaks the semantic security with a non-negligible probability.

Security Reduction: the Recipe

Proposition (Security Reduction)

Suppose we have a cryptosystem $\mathcal{E}$ and we believe the problem $\mathcal{X}$ is computationally hard. To prove that $\mathcal{E}$ is secure based on $\mathcal{X}$:

1: Assume there exists an efficient adversary $\mathcal{A}$ who breaks $\mathcal{E}$ with a non-negligible probability $\varepsilon(\lambda)$.

2: Construct an efficient algorithm $\mathcal{B}$ (the **reduction**) that solves $\mathcal{X}$ using oracle access to $\mathcal{A}$: on input x of the problem $\mathcal{X}$, $\mathcal{B}$ generates the input to the security game of $\mathcal{E}$, runs $\mathcal{A}$ and uses its output to solve $\mathcal{X}$.

3: If $\mathcal{B}$ is constructed properly, it solves $\mathcal{X}$ with probability at least $\varepsilon(\lambda)/\lambda^{\gamma}$ for some γ, which is non-negligible.

4: Hence assumption $\mathcal{X}$ implies security of $\mathcal{E}$.

Discrete Logarithm Assumption (DL)

Definition

Assume that $\mathbb{G}$ is a cyclic group of prime order $r = \Theta(2^\lambda)$ generated by $g \in \mathbb{G}$. Define the following game:

1: Both Ch and $\mathcal{A}$ take a description $\mathbb{G}$ as an input: order r and generator $g \in \mathbb{G}$.

2: Ch computes $\alpha \leftarrow\$ \mathbb{Z}_r, u \leftarrow g^\alpha$ and sends $u \in \mathbb{G}$ to $\mathcal{A}$.

3: The adversary $\mathcal{A}$ outputs $\widehat{\alpha} \in \mathbb{Z}_r$.

We define $\mathcal{A}$'s **advantage in solving the discrete logarithm problem in** $\mathbb{G}$, denoted as $\mathrm{Adv}^{\mathrm{DL}}_{\mathcal{A},\mathbb{G}}$, as the probability that $\widehat{\alpha} = \alpha$.

Definition

The **Discrete Logarithm Assumption** holds in the group $\mathbb{G}$ if for any efficient adversary $\mathcal{A}$ the advantage $\mathrm{Adv}^{\mathrm{DL}}_{\mathcal{A},\mathbb{G}}(\lambda)$ is negligible in λ.

Computational Diffie-Hellman Assumption (CDH)

Definition

Let $\mathbb{G}$ be a cyclic group of prime order $r = \Theta(2^\lambda)$ generated by $g \in \mathbb{G}$. Define the following game:

1: Ch computes $\alpha, \beta \leftarrow\$ \mathbb{Z}_r$, $u \leftarrow g^\alpha, v \leftarrow g^\beta, w \leftarrow g^{\alpha\beta}$ and sends $u, v \in \mathbb{G}$ to $\mathcal{A}$.

2: The adversary $\mathcal{A}$ outputs $\widehat{w} \in \mathbb{G}$.

We define $\mathcal{A}$'s **advantage in solving the computational Diffie-Hellman problem in** $\mathbb{G}$ as $\mathsf{Adv}^{\mathsf{CDH}}_{\mathcal{A},\mathbb{G}}(\lambda) := \Pr[\widehat{w} = w]$.

Definition

The **Computational Diffie-Hellman Assumption** holds in the group $\mathbb{G}$ if for any efficient adversary $\mathcal{A}$, $\mathsf{Adv}^{\mathsf{CDH}}_{\mathcal{A},\mathbb{G}}(\lambda) = \mathsf{negl}(\lambda)$.

Decisional Diffie-Hellman Assumption (DDH)

Definition

Let $\mathbb{G}$ be a cyclic group of prime order $r = \Theta(2^\lambda)$ generated by $g \in \mathbb{G}$. Define the following game:

1: $\mathcal{C}h$ computes $\alpha, \beta, \gamma \leftarrow\$ \mathbb{Z}_r$, $u \leftarrow g^\alpha, v \leftarrow g^\beta, w_0 \leftarrow g^{\alpha\beta}, w_1 \leftarrow g^\gamma$. Then, $\mathcal{C}h$ flips a coin $b \leftarrow\$ \{0,1\}$ and sends (u, v, w_b) to $\mathcal{A}$.

2: The adversary $\mathcal{A}$ outputs the predicted bit $\widehat{b} \in \{0,1\}$.

We define $\mathcal{A}$'s **advantage in solving the Decisional Diffie-Hellman problem in** $\mathbb{G}$ as

$$\mathsf{Adv}^{\mathsf{DDH}}_{\mathcal{A},\mathbb{G}}(\lambda) := \left| \Pr[b = \widehat{b}] - \frac{1}{2} \right|.$$

DDH is considerably stronger (and more fragile) than CDH. Let us show when DDH does not hold!

Even-ordered Cyclic Group

Theorem

Suppose that $\mathbb{G}$ is a cyclic group of an even order. Then, the Decisional Diffie-Hellman Assumption does not hold in $\mathbb{G}$. In fact, there is an efficient adversary $\mathcal{A}$ with an advantage $1/4$.

Idea of proof. Let $\#\mathbb{G} = 2n$. We first prove the following statement:

Lemma

Suppose $u = g^{\alpha}$. Then, α is even if and only if $u^n = 1$.

Lemma proof. If $\alpha = 2\alpha'$, then

$$u^n = g^{\alpha n} = g^{2\alpha' n} = (g^{2n})^{\alpha'} = 1^{\alpha'} = 1.$$

Therefore, if $u = g^{\alpha}, v = g^{\beta}, w = g^{\gamma}$, the adversary knows the parity of α, β, γ. He then checks if parity of $\alpha\beta$ equals parity of γ.

Polynomials

Definition

Definition

A **polynomial** $p(T)$ is a function of the form

$$p(T) = c_0 + c_1 T + c_2 T^2 + \cdots + c_n T^n = \sum_{k=0}^{n} c_k T^k,$$

where $c_0, c_1, \ldots, c_n$ are coefficients of the polynomial.

Definition

A set of polynomials depending on T with coefficients in a field $\mathbb{F}$ is denoted as $\mathbb{F}[T]$, that is

$$\mathbb{F}[T] = \left\{ p(T) = \sum_{k=0}^{n} c_k T^k : c_k \in \mathbb{F},\ k = 0, \ldots, n \right\}.$$

Examples of Polynomials

Example

Consider the finite field $\mathbb{F}_3$. Then, some examples of polynomials from $\mathbb{F}_3[T]$ are listed below:

1. $p(T) = 1 + T + 2T^2$.
2. $q(T) = 1 + T^2 + T^3$.
3. $r(T) = 2T^3$.

If we were to evaluate these polynomials at $1 \in \mathbb{F}_3$, we would get:

1. $p(1) = 1 + 1 + 2 \cdot 1 \bmod 3 = 1$.
2. $q(1) = 1 + 1 + 1 \bmod 3 = 0$.
3. $r(1) = 2 \cdot 1 = 2$.

More about polynomials

Definition

The **degree** of a polynomial $p(T) = c_0 + c_1 T + c_2 T^2 + \dots$ is the largest $k \in \mathbb{Z}_{\geq 0}$ such that $c_k \neq 0$. We denote the degree of a polynomial as $\deg p$. We also denote $\mathbb{F}[T]^{\leq m} := \{p \in \mathbb{F}[T] : \deg p \leq m\}$.

Example

The degree of the polynomial $p(T) = 1 + 2T + T^2$ is 2, so $p \in \mathbb{F}_3[T]^{\leq 2}$.

Theorem

For any polynomials $p, q \in \mathbb{F}[T]$ and $n := \deg p$, $m := \deg q$, the following two statements are true:

1. $\deg(pq) = n + m$.
2. $\deg(p + q) = \max\{n, m\}$ if $n \neq m$, and $\deg(p + q) \leq m$ if $m = n$.

Roots of Polynomials

Definition

Let $p \in \mathbb{F}[T]$ be a polynomial of degree $\deg p \geq 1$. A field element $\alpha \in \mathbb{F}$ is called a **root** of p if $p(\alpha) = 0$.

Example

Consider the polynomial $p(T) = 1 + T + T^2 \in \mathbb{F}_3[T]$. Then, $\alpha = 1$ is a root of p since $p(\alpha) = 1 + 1 + 1 \bmod 3 = 0$.

Theorem

Let $p \in \mathbb{F}[T]$, $\deg p \geq 1$. Then, $\alpha \in \mathbb{F}$ is a root of p if and only if there exists a polynomial q (with $\deg q = \deg p - 1$) such that

$$p(T) = (T - \alpha)q(T).$$

Polynomial Division

Theorem

Given $f, g \in \mathbb{F}[T]$ with $g \neq 0$, there are unique polynomials $q, r \in \mathbb{F}[T]$ such that

$$f = q \cdot g + r,\ 0 \leq \deg r < \deg g.$$

Example

Consider $f(T) = T^3 + 2$ and $g(T) = T + 1$ over $\mathbb{R}$. Then, we can write $f(T) = (T^2 - T + 1)g(T) + 1$, so the remainder of the division is $r \equiv 1$. Typically, we denote this as:

$$f \operatorname{div} g = T^2 - T + 1, \quad f \bmod g = 1.$$

The notation is pretty similar to one used in integer division.

Polynomial Divisibility

Definition

A polynomial $f \in \mathbb{F}[T]$ is called **divisible** by $g \in \mathbb{F}[T]$ (or, g **divides** f, written as $g \mid f$) if there exists $h \in \mathbb{F}[T]$ such that $f = gh$.

Theorem

If $\alpha \in \mathbb{F}$ is a root of $p \in \mathbb{F}[T]$, then $(T - \alpha) \mid p$.

Definition

A polynomial $f \in \mathbb{F}[T]$ is said to be **irreducible** in $\mathbb{F}$ if there are no polynomials $g, h \in \mathbb{F}[T]$ both of positive degree such that $f = gh$.

Polynomial Divisibility

Example

A polynomial $f(T) = T^2 + 16$ is irreducible in $\mathbb{R}$. Also $f(T) = T^2 - 2$ is irreducible over $\mathbb{Q}$, yet it is reducible over $\mathbb{R}$:
$f(T) = (T - \sqrt{2})(T + \sqrt{2})$.

Example

There are no polynomials over complex numbers $\mathbb{C}$ with degree more than 1 that are irreducible. This follows from the *fundamental theorem of algebra*. For example, $T^2 + 16 = (T - 4i)(T + 4i)$.

Interpolation

Question

How can we define the polynomial?

The most obvious way is to specify coefficients $(c_0, c_1, \ldots, c_n)$. Can we do it in a different way?

Theorem

Given $n+1$ distinct points $(x_0, y_0), \ldots, (x_n, y_n)$, there exists a unique polynomial p of degree at most n such that $p(x_i) = y_i$ for all $i = 0, \ldots, n$.

Illustration with two points

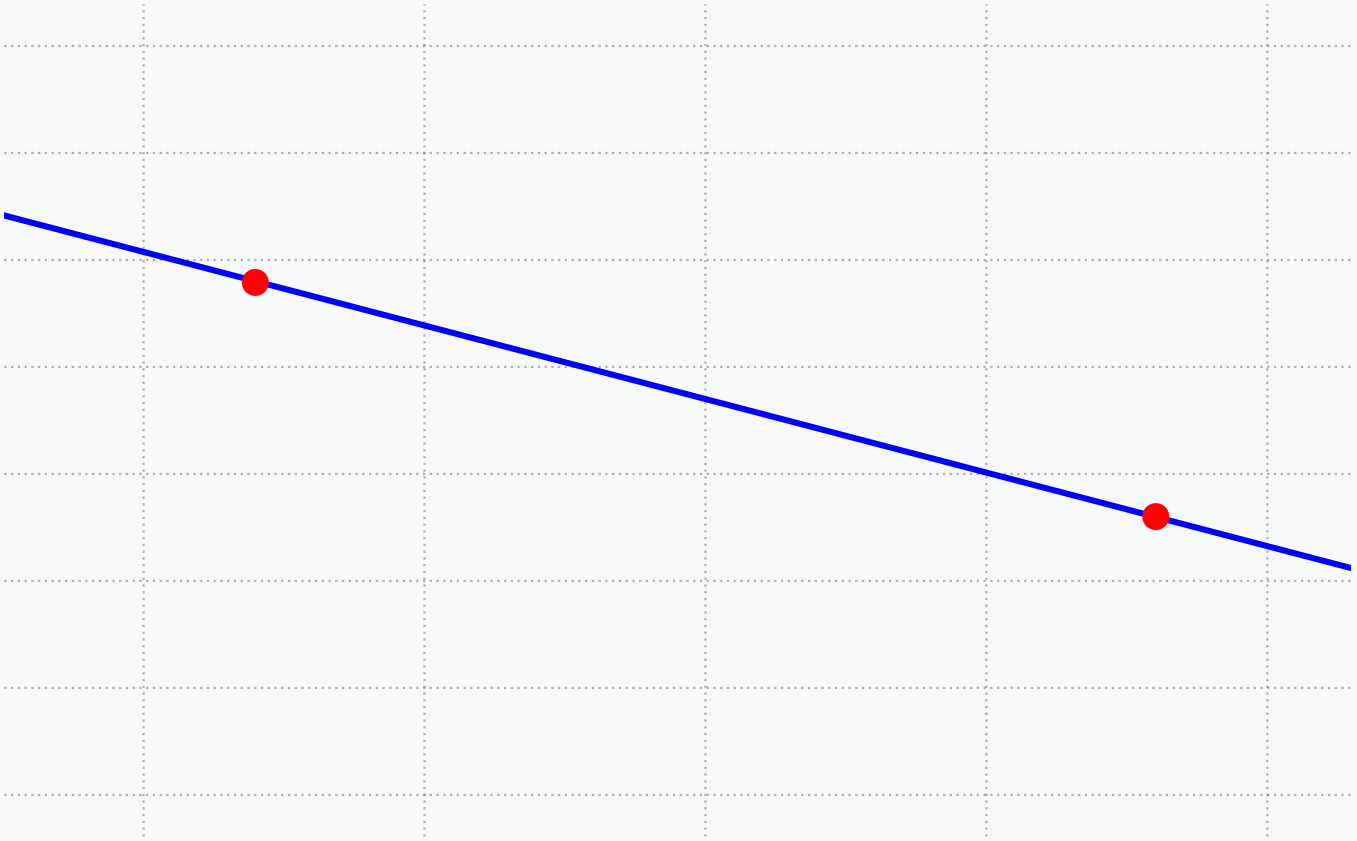


Figure: 2 points on the plane uniquely define the polynomial of degree 1 (linear function).

Illustration with five points

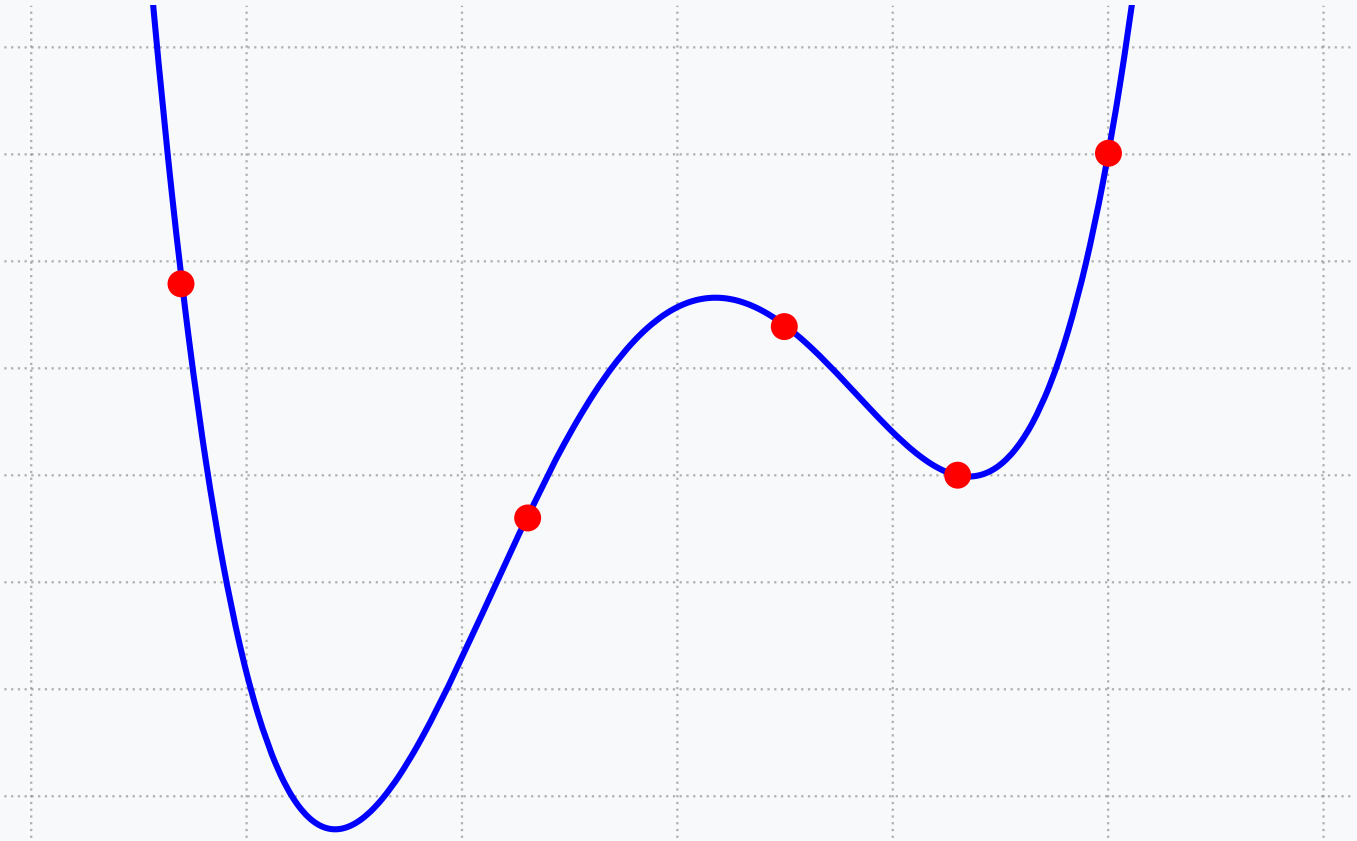


Figure: 5 points on the plane uniquely define the polynomial of degree 4.

Lagrange Interpolation

One of the ways to interpolate the polynomial is to use the Lagrange interpolation.

Theorem

Given $n+1$ distinct points $(x_0, y_0), \ldots, (x_n, y_n)$, the unique polynomial L of degree at most n that passes through these points is given by

$$L(T) = \sum_{i=0}^{n} y_i \ell_i(T), \quad \ell_i(T) = \prod_{j=0, j\neq i}^{n} \frac{T - x_j}{x_i - x_j}.$$

Example

For points $(0, 1)$ and $(1, 2)$, we get

$$L(T) = 1 \cdot \frac{T-1}{0-1} + 2 \cdot \frac{T-0}{1-0} = T + 1.$$

Application: Shamir Secret Sharing

Motivation

How to share a secret α among n people in such a way that any t of them can reconstruct the secret, but any $t-1$ cannot?

Definition

Secret Sharing scheme is a pair of efficient algorithms $(\mathrm{Gen}, \mathrm{Comb})$ which work as follows:

- $\mathrm{Gen}(\alpha, t, n)$: probabilistic sharing algorithm that yields n shards $(\alpha_1, \ldots, \alpha_n)$ for which t shards are needed to reconstruct the secret α.
- $\mathrm{Comb}(\mathcal{I}, \{\alpha_i\}_{i \in \mathcal{I}})$: deterministic reconstruction algorithm that reconstructs the secret α from the shards $\mathcal{I} \subset \{1, \ldots, n\}$ of size t.

Shamir's Protocol

Correctness

For every $\alpha \in F$, for every possible output $(\alpha_1, \ldots, \alpha_n) \leftarrow \mathsf{Gen}(\alpha, t, n)$, and any t-size subset $\mathcal{I}$ of $\{1, \ldots, n\}$ we require

$$\mathsf{Comb}(\mathcal{I}, \{\alpha_i\}_{i \in \mathcal{I}}) = \alpha.$$

Definition

Shamir's protocol works over $F = \mathbb{F}_q$ as follows:

- $\mathsf{Gen}(\alpha, t, n)$: choose random $k_1, \ldots, k_{t-1} \leftarrow\!\!\$\ \mathbb{F}_q$ and define the polynomial

$$\omega(T) := \alpha + k_1 T + k_2 T^2 + \cdots + k_{t-1} T^{t-1} \in \mathbb{F}_q[T]^{<t},$$

  and then compute $\alpha_i \leftarrow \omega(i) \in \mathbb{F}_q$ for each $i \in \{1, \ldots, n\}$.

- $\mathsf{Comb}(\mathcal{I}, \{\alpha_i\}_{i \in \mathcal{I}})$: interpolate the polynomial ω using the Lagrange interpolation and output $\omega(0) = \alpha$.

Shamir's Protocol: Intuition

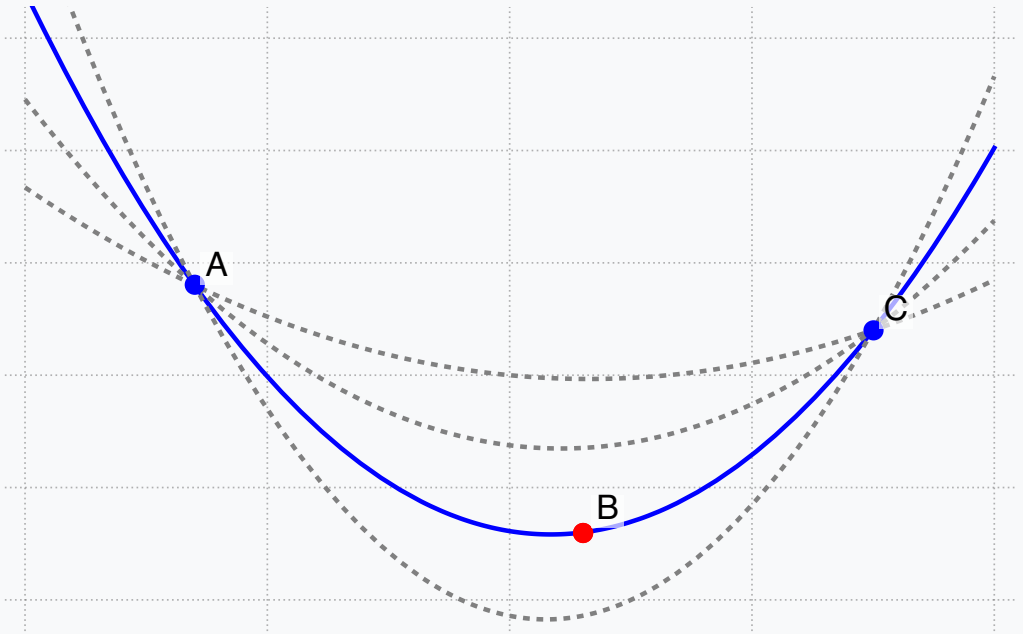


Figure: There are infinitely many quadratic polynomials passing through two blue points (gray dashed lines). However, knowing the red point allows us to uniquely determine the polynomial and thus get its value at 0.

Reed-Solomon Codes

Reed-Solomon codes

Reed-Solomon codes are an error-correcting algorithm based on polynomials. They allow restoring lost or corrupted data, implementing threshold secret sharing, and they are used in some ZK protocols (we will come back to them when building STARKs).

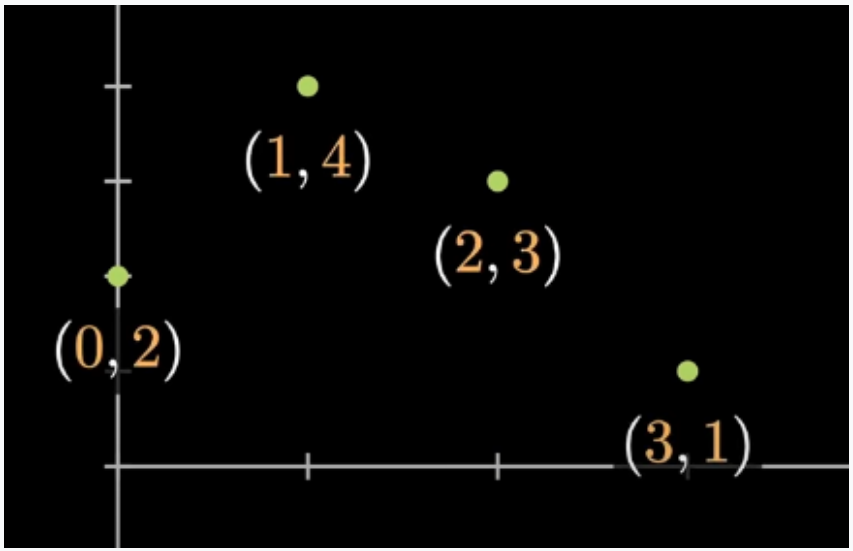


Figure: Polynomial of degree n is uniquely defined by $n + 1$ points. Defining more points on the same polynomial adds a redundancy, which can be used to restore the polynomial even if some points are missing.

Schwartz-Zippel Lemma

Lemma (Single Variable Schwartz-Zippel Lemma)

Let $\mathbb{F}$ be a finite field and let $f \in \mathbb{F}[T]$ be a non-zero polynomial. Then,

$$\Pr_{\alpha \leftarrow \$ \mathbb{F}}[f(\alpha) = 0] \leq \frac{\deg f}{\#\mathbb{F}}.$$

Proof Idea. The polynomial of degree $n := \deg f$ has at most n roots. Therefore, when we pick α uniformly at random from $\mathbb{F}$, there is at most $n/\#\mathbb{F}$ chance to pick a root.

Lemma (Multivariable Schwartz-Zippel)

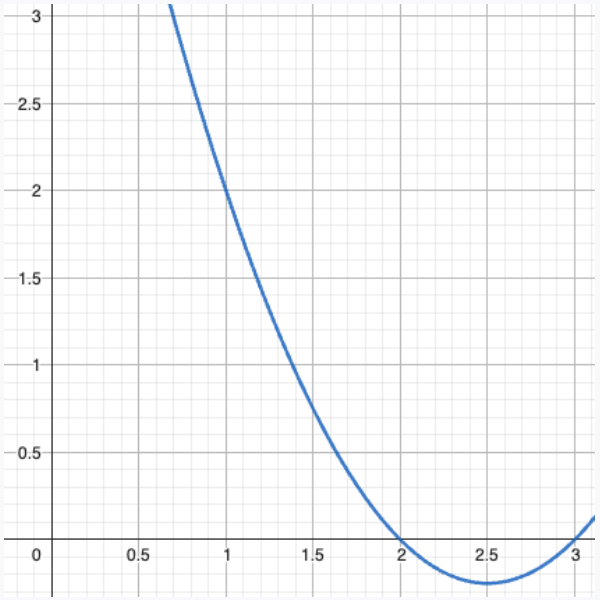
Let $S \subseteq \mathbb{F}^n$ be finite and f be a non-zero polynomial in n variables. Then,

$$\Pr_{\alpha \leftarrow \$ S}[f(\alpha) = 0] \leq \frac{\deg f}{\#S}.$$

Schwartz-Zippel Lemma: Example

Example

Suppose we are working over the Mersenne prime field of order $p = 2^{31} - 1$ with a polynomial $f(T, W) = T^{50} + 3TW + W^5$. If we pick (α, β) at random from $\mathbb{F}_p^2$, the probability of $f(\alpha, \beta) = 0$ is at most $50/(2^{31} - 1)$, providing approximately 25 bits of security.



Polynomial Equality Test

Corollary (Polynomial Equality Test)

Suppose P and Q are two distinct polynomials with a maximum degree d over the finite field $\mathbb{F}$. Then,

$$\Pr_{r \leftarrow\$ \mathbb{F}}[P(r) = Q(r)] \leq \frac{d}{\#\mathbb{F}}.$$

As a consequence, for large fields, the probability of two distinct polynomials being equal at a random point is negligible.

Proof

Let $Z(T) := P(T) - Q(T)$. Of course, $\deg Z$ is at most d. Also note that $Z(r) = 0$ iff $P(r) = Q(r)$. Applying Schwartz-Zippel lemma, the probability of $Z(r) = 0$ for $r \leftarrow\$ \mathbb{F}$ is bounded above by $d/\#\mathbb{F}$. □

Thank you for your attention



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